

Lite Arm SDK: Repository Structure and API

- Language: C++20

Repository Structure

```
lite_arm_sdk/  
|-- CMakeLists.txt  
|-- include  
|   |-- lite_arm.h           (main SDK class API)  
|   |-- configurator.h      (configuration API)  
|   |-- kinematics.h        (kinematics API)  
|   `-- types.h             (shared data types/enums)  
|-- examples/  
|   |-- manual_mode.cpp  
|   |-- teach_mode.cpp  
|   |-- playback_mode.cpp  
|   `-- calculate_kinematics.cpp
```

Core SDK API (LiteArm)

API	Description	Group
<code>LiteArm(local_ip, local_port)</code>	Create SDK client and initialize configurator transport.	constructor
<code>start(), stop()</code>	Start or stop arm runtime session.	connection
<code>enableJoint(state), disableJoint(state)</code>	Enable or disable arm/gripper joints by boolean state arrays.	connection
<code>setInterpolationMethod(method)</code>	Select interpolation strategy for motion control.	motion mode
<code>movePos(jointPos, v=50, t=0)</code>	Command joint-space target position.	manual motion
<code>movePosPath(path, vPath, tPath)</code>	Command sequence of joint-space waypoints.	manual motion
<code>moveVel(jointVel, a=10, t=0)</code>	Command joint-space velocity.	manual motion
<code>moveEEPoint(pose, v=50, t=0)</code>	Move end-effector to cartesian pose (point mode).	manual motion
<code>moveEEPointPath(path, vPath, tPath)</code>	Move through end-effector pose waypoints (point mode).	manual motion

API	Description	Group
<code>moveEELine(pose, v=50, t=0)</code>	Move end-effector to cartesian pose (line mode).	manual motion
<code>moveEELinePath(path, vPath, tPath)</code>	Move through end-effector pose waypoints (line mode).	manual motion
<code>moveToolPoint(pose, v=50, t=0)</code>	Move tool frame to cartesian pose (point mode).	manual motion
<code>moveToolPointPath(path, vPath, tPath)</code>	Move tool frame through cartesian waypoints (point mode).	manual motion
<code>moveToolLine(pose, v=50, t=0)</code>	Move tool frame to cartesian pose (line mode).	manual motion
<code>moveToolLinePath(path, vPath, tPath)</code>	Move tool frame through cartesian waypoints (line mode).	manual motion
<code>goHome(v=50)</code>	Move robot arm to predefined home pose.	manual motion
<code>moveGripperPos(pos, t=0)</code>	Set gripper joint position.	manual motion
<code>startTeach(), stopTeach()</code>	Enter or exit teach mode.	teach mode
<code>startPlayback(), stopPlayback(), resetPlayback()</code>	Control playback mode lifecycle.	playback mode
<code>loadPlayback(filename)</code>	Load playback trajectory or script file.	playback mode
<code>getPos(), getVel(), getTor()</code>	Read current joint position, velocity, and torque.	feedback
<code>getLastCommand()</code>	Read last command state issued to robot.	feedback
<code>getEEPose(), getToolPose()</code>	Read current end-effector/tool cartesian pose.	feedback
<code>getStatus(), getErrorCode(), getWarningCode()</code>	Read high-level runtime status and fault indicators.	status
<code>getJointStatus()</code>	Read per-joint enable/error status for arm and gripper.	status
<code>getConfigurator(), getKinematics()</code>	Access specialized subsystems for config/kinematics.	subsystems

Configuration API (`Configurator`)

API	Description
<code>setToolOffset(offset)</code>	Set tool frame offset from end-effector frame.
<code>getToolOffset()</code>	Get configured tool frame offset.
<code>setArmPosLimits(limit), setArmVelLimits(limit)</code>	Set joint position/velocity safety limits.
<code>getArmPosLimits(), getArmVelLimits()</code>	Get active joint safety limits.
<code>getArmMaxPos(), getArmMaxVel()</code>	Get hardware max position/velocity constraints.
<code>getArmMaxPosJump(), getArmMaxVelJump()</code>	Get anti-jump protection limits.
<code>getArmMaxPosFollowError(), getArmMaxVelFollowError(), getArmMaxTorFollowError()</code>	Get following-error thresholds.
<code>setGripperPosLimits(limit), getGripperPosLimits()</code>	Set or get gripper position limits.
<code>getGripperMaxPos()</code>	Get gripper hardware max range.

Kinematics API (`Kinematics`)

API	Description
<code>forwardKinematics(arm_pos)</code>	Compute cartesian pose from 6-joint arm state.
<code>inverseKinematics(tool_pose, arm_pos)</code>	Solve 6-joint arm state for a target cartesian pose.
<code>eulerToQuaternion(euler)</code>	Convert intrinsic XYZ Euler angles to quaternion.
<code>quaternionToEuler(quat)</code>	Convert quaternion to intrinsic XYZ Euler angles.
<code>eeToToolPose(ee_pose, tool_offset)</code>	Transform end-effector pose into tool pose.
<code>toolToEEPose(tool_pose, tool_offset)</code>	Transform tool pose into end-effector pose.

Types (`types.h`)

Type / Enum	Description
<code>Pose</code>	Cartesian pose = <code>Position</code> + <code>Quaternion</code> orientation.
<code>JointState6f / JointState1f</code>	6-DOF arm joint values / 1-DOF gripper value.
<code>RobotJointStatef / RobotJointStateb</code>	Combined arm+grripper state in float or bool forms.
<code>JointLimits6 / JointLimits1</code>	Per-joint min/max limits for arm and gripper.

Type / Enum	Description
Status	kInitialized, kStarted, kWarning, kError, kRecovery.
WarningCode	kNone, kCmdLimitSaturation, kCmdJumpSaturation.
ErrorCode	kNone, kConnectionFailed, kEnableFailed, and limit/following errors.
RobotJointStatus	Per-joint status arrays (kInitialized/kEnabled/kDisabled/kError).